
Self-Directed Decomposition Empowers Reasoning Potentials in Large Language Models

Abstract

1 Large Language Models (LLMs) have demonstrated remarkable advancements
2 in natural language processing and reasoning tasks, yet often struggle with logi-
3 cal coherence during problem-solving. This paper introduces Self-Directed De-
4 composition (SD), a novel prompting strategy enabling LLMs to autonomously
5 decompose reasoning problems into manageable sub-tasks without human inter-
6 vention, allowing models to determine their own approach with adaptive flexibility
7 across diverse reasoning domains. Experiments across seven reasoning tasks reveal
8 that this methodology particularly enhances performance on deductive, inductive,
9 mathematical, commonsense, and scientific reasoning tasks, while showing more
10 modest benefits for abductive and causal reasoning tasks, achieving 62.26% overall
11 median accuracy compared to 49.64% and 46.43% for zero-shot and zero-shot
12 Chain-of-Thought (CoT) approaches, respectively. Error and statistical analy-
13 sis demonstrates that SD significantly transforms reasoning patterns by reducing
14 wrong selection errors but increasing process mistakes for simpler variants, with
15 only SD1 maintaining optimal balance. We discover a counterintuitive negative
16 correlation between token consumption and accuracy ($R^2 = 0.162, p = 0.004$),
17 challenging conventional resource-performance assumptions. Abductive reasoning
18 demonstrates critical vulnerability to decomposition strategies, showing significant
19 perspective errors increase ($R^2 = 0.66$). These findings explain why SD1 outper-
20 forms other variants: it balances different error types effectively while avoiding the
21 complexity-accuracy trade-off that affects simpler decomposition strategies.

22 1 Introduction

23 Large Language Models (LLMs) have recently made significant progress in natural language pro-
24 cessing (NLP), decision-making, and cognitive learning tasks (16; 18). During the LLMs evolution,
25 zero-shot Chain-of-Thought (CoT) is a widely used prompt engineering technique to enhance LLM
26 efficiency by instructing the model with “*Let’s think step-by-step*” to break down tasks into intermedi-
27 ate steps (13; 30). This approach addresses complex tasks that cannot be resolved through single-step
28 procedures into smaller, tractable step-by-step components. A significant advantage of this strategy
29 is that it operates without necessitating pre-defined exemplars or explicit instructional guidance in
30 the prompts. However, the main limitations of this strategy are that LLMs can still make semantic
31 misunderstanding errors or skip intermediate steps during reasoning (24), consequently yielding
32 erroneous conclusions and limiting their generalization capabilities.
33 Afterwards, based on the CoT’s success, various prompting techniques have been developed
34 to further enhance LLMs’ reasoning capabilities. Including Least-to-Most prompting(36), self-
35 consistency(26), Logical CoT (Logi-CoT)(15), Tree of Thoughts (ToT)(35), Thread of Thought
36 (ThoT)(37), Chain of Table(29), System 2 Attention (S2A)(33), Graph of Thoughts (GoT)(2), Task
37 Dynamic Decomposition(28). However, these approaches typically rely on predefined exemplars

or researcher-designed algorithms that constrain the model’s autonomy, rather than allowing it to independently select the best analytical approaches. Researchers have also explored emotional prompting strategies that engage LLMs through emotional cues(14). Most emotional prompts require specific social contexts or additional psychological cues, except for emotional stimuli EP02, “*This is very important to my career*”, which stands out as more generic.

To enhance the autonomous functioning of LLMs, this research introduces a novel prompting template, Self-Directed Decomposition (SD), which facilitates the resolution of diverse tasks without extraneous guidance. This approach enables LLMs to systematically decompose complex problems into constituent, more tractable sub-tasks based on their internal representations. To evaluate the effectiveness of SD strategies, we compare them with zero-shot, zero-shot CoT, and emotional stimuli EP02 as baseline prompts across seven reasoning datasets.

Through comprehensive experiments across multiple reasoning domains, we analyze how different SD formulations affect performance and error patterns, providing insights into the mechanisms underlying autonomous decomposition in LLMs. The remainder of this paper is structured as follows: Section 2 outlines the definitions of SD strategies. Section 3 presents experimental setups across the different datasets of reasoning tasks and the evaluation standards for the mistakes made in the processes. Section 4 shows experimental results for different prompting strategies and discusses the main reason why SD strategies perform differently with error and statistical analysis. Finally, Sections 5 and 6 summarize the main limitations and conclude this approach’s implications for future research and applications and beyond.

2 Self-Directed Decomposition Prompting Strategies

This paper developed a series of prompting techniques called Self-Directed Decomposition (SD). These SD prompts enable LLMs to autonomously decompose complex reasoning problems into manageable sub-tasks, with varying levels of structural guidance to determine optimal decomposition complexity. SD1 includes explicit decomposition instructions using modal operators (“*should*”) and structured terminology (“*sub-tasks*”), providing comprehensive guidance for the decomposition process. SD2 shortens the instruction while maintaining the core decomposition directive. SD3 adds a politeness marker (“*please*”), which may affect model behavior similar to emotional stimuli(14). SD4 adopts a minimal wording approach(9; 22; 38).

Self-Directed Decomposition 1 *When you deal with such problems, you should do a problem decomposition like sub-tasks to analyse the problem.*

Self-Directed Decomposition 2 *When you deal with a problem, decompose the problem and solve the task.*

Self-Directed Decomposition 3 *Please decompose the problem and solve it.*

Self-Directed Decomposition 4 *Decompose and solve.*

Given facts: Lily is a lion. Brian is a frog. Julius is a rhino. Julius is white. Greg is a swan. Bernhard is a lion. Bernhard is grey. Brian is white. Greg is white. Based on the given facts above, do a reasonable inference on this question using inductive reasoning: What colour is Lily?		
Zero-Shot CoT: Based on the given facts, we know that Lily is a lion. We also know that Bernhard is a lion and is grey, and Brian is a frog and is white. Since there is no direct information about the color of lions in the provided facts, we can make an inference based on the colors of the other animals mentioned. ✖ Given that Brian, a frog, is white, and Bernhard, a lion, is grey, we can infer that the color of an animal does not necessarily correspond to its species. Therefore, we cannot definitively determine the color of Lily based on the information provided. ✖ In this case, we cannot make a reasonable inference about the color of Lily based on the given facts using inductive reasoning. ✖	Emotional Stimuli (EP02): Based on the given facts and using inductive reasoning, we can infer that Lily, being a lion, could potentially be either grey or white. ✖ Both Bernhard and Brian, who are lions like Lily, are described as grey and white respectively. ✖ Since there are examples of lions in both grey and white colors in the given facts, we cannot definitively determine the color of Lily without additional information. ✖	Self-Directed Decomposition: To determine the color of Lily, we can break down the problem into the following sub-tasks based on the given facts: 1. Lily is a lion. ✓ 2. Bernhard is a lion and is grey. ✓ 3. Since Lily is a lion and Bernhard is a lion, it is reasonable to infer that lions in this context are grey. ✓ 4. Therefore, based on the given information, it is reasonable to infer that Lily, being a lion, is grey. ✓ So, based on the inductive reasoning and the information provided, the reasonable inference is that Lily is grey. ✓

Figure 1: Self-directed decomposition solving an inductive reasoning task based on the model’s understanding.

This investigation focuses on understanding how different levels of decomposition guidance affect reasoning performance and error patterns across various reasoning domains. SD1-SD4 variants test whether detailed decomposition instructions improve performance or create procedural complexity. Each variant progressively simplifies the instruction structure. Figure 1 illustrates the decomposition approach using an inductive reasoning task as an example. This figure shows how the method systematically breaks down complex problems while maintaining logical coherence throughout the reasoning process.

3 Experiments

3.1 Datasets

These experiments aim to measure how SD strategies affect LLM performance compared to existing techniques, quantify their impact on error patterns, and identify which reasoning tasks benefit most from autonomous decomposition. The reasoning categories and datasets include: (1) **Deductive Reasoning**: A top-down approach. Starting with a general principle or assumption to conclude specific answers (8). **Datasets**: bAbI (task 15) and EntailmentBank (E-Bank) (5; 32); (2) **Inductive Reasoning**: Derives general principles based on the given observations (7; 10). **Datasets**: CLUTRR and bAbI (task 16) (21; 32); (3) **Mathematical Reasoning**: Using mathematical or logical principles to find the solution to problems (20). **Datasets**: Mathematics (Math) and SVAMP (17; 19); (4) **Scientific Reasoning**: Similar to mathematical reasoning, but using a set of collegiate-level scientific problems from calculus, chemistry and physics domains. **Dataset**: Scibench (27). (5) **Commonsense Reasoning**: Based on humans’ daily knowledge and shared understanding, interpret and navigate rules and situations (1). **Dataset**: CommonsenseQA (CommonQA), PiQA and Pep-3K (4; 23; 25); (6) **Abductive Reasoning**: Infers the most likely explanation for the given observation, focusing on plausible hypotheses rather than deriving necessarily true conclusions (31). **Datasets**: α NLI and ART (3; 7); and (7) **Causal Reasoning**: Focuses on identifying and understanding the cause-and-effect relationships among the given facts or observations (11). **Datasets**: E-care and Balanced-COPA (B-COPA)(6; 12).

3.2 Implementation

This paper used the GPT-3.5-turbo model via APIs as the backbone model for conducting the above datasets. The purpose of using this model rather than larger parameter models such as the GPT-4o series is to demonstrate that effective reasoning can be achieved through improved prompting strategies rather than simply increasing model size. This approach emphasizes that empowering a model’s potential through strategic prompting may be more efficient than relying on larger architectures to overcome reasoning limitations. Each API call is dedicated to testing exactly one prompting strategy (SDs, zero-shot, zero-shot CoT, or emotional EP02) to avoid cross-contamination of information. The temperature is set to 0 throughout the experiment. From each dataset, a representative sample comprising 10 tasks was manually extracted through random selection procedures (with an expanded sample of 40 tasks from the Scibench dataset), prioritizing methodological feasibility and assessment efficiency for generalization capabilities rather than task-specific optimization paradigms. Throughout the experimental procedures, each task was presented precisely once per prompting strategy to maintain methodological consistency. Accuracy was determined based on whether the response from LLMs provided the correct answer. All statistical procedures were implemented using Python’s SciPy and statsmodels libraries, with visualization support from Matplotlib and Seaborn packages. In all box plots presented in this paper, the box boundaries represent the first (**Q1**) and third (**Q3**) quartiles, the middle line shows the median (**Q2**), whiskers extend to the most extreme data points within 1.5 times the interquartile range (**IQR**) from the box boundaries, and individual points beyond the whiskers indicate outliers. All the tokens were calculated via OpenAI Tokenizer.

3.3 Evaluation Metrics

3.3.1 Error Analysis

To facilitate a more comprehensive evaluation of the performance parameters and inherent limitations of the proposed methodology, this investigation implements a systematic error analysis framework

predicated on the five-category error taxonomy established by Xu et al. (34). This error analysis is to reveal how SD strategies influence the model’s reasoning process and provides insights into the underlying mechanisms through which SD affects model reasoning. The error taxonomy categorizes mistakes into five distinct types: (1) **Wrong Selection (WS)**: At the beginning of the reasoning process, LLMs choose the wrong facts or ignore the necessary facts to generate the answer. (2) **Hallucination (HA)**: LLMs incorporate information elements or factual assertions that lack empirical verification or explicitly contradict the established contexts. (3) **No Reasoning (NR)**: LLMs listed the given facts without reasoning from the facts to conclude an answer. (4) **Perspective Mistake (PPM)**: LLMs started from an incorrect or irrelevant point of view to process the answer. (5) **Process Mistake (PM)**: LLMs started from the correct perspective but made mistakes during reasoning.

3.3.2 Statistical Analysis

To establish statistically rigorous evaluation protocols for the accuracy and error analysis, this investigation employs multiple complementary statistical frameworks that quantify the relationships between dependent variables (error proportions, accuracy) and independent variables (prompting methods). The primary statistical metrics utilized in this investigation include: **Coefficient of Determination (R^2)**: The explanatory power of linear regression models is quantified using R^2 values, which measure the proportion of variance in error rates attributable to the independent variables. Higher R^2 values (approaching 1.0) indicate stronger predictive relationships, while values closer to zero indicate minimal association. **Statistical Significance (p -value)**: All reported correlations and regression models are subjected to hypothesis testing with a significance threshold of $\alpha = 0.05$. The p -values reported alongside regression analyses indicate the probability that observed relationships could occur by random chance, with lower values conferring higher confidence in the validity of the findings. **Regression Coefficients**: Linear regression equations provide quantitative measures of effect size, where the slope coefficient represents the expected unit change in error proportion per unit increase in the independent variable. These coefficients facilitate direct comparisons of effect magnitudes across different error types and reasoning categories. **Distributional Metrics**: Box plots visualize the central tendency (median), dispersion (interquartile range), and outlier characteristics of error distributions across methods and reasoning types. The positional differences in median lines and box heights quantify both the absolute performance differences and the variability in performance across experimental conditions. **ANOVA**: Analysis of variance tests determine whether observed differences in error rates across categorical variables (methods, reasoning types) reflect statistically significant patterns rather than random variation. This framework enables systematic comparisons across multiple experimental conditions while controlling for familywise error rates.

4 Results & Discussion

4.1 SD1 Achieves Superior Decomposition Performance with Optimal Error Balance

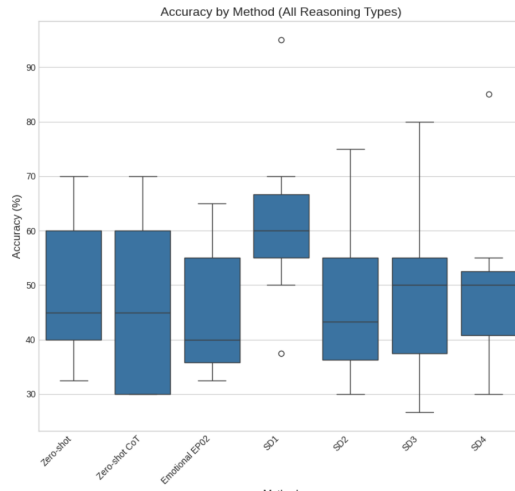


Figure 2: Overall reasoning accuracy vs. prompting methods.

As shown in Figure 2, SD1’s structured decomposition approach eliminates the trade-off between accuracy and consistency that plagues other methods, achieving both superior performance (62.26%, $p < 0.05$) and tight error bounds (50%-70% IQR). This dual advantage stems from explicit decomposition instructions using modal operators ("should") and structured terminology ("sub-tasks"), which provide sufficient guidance without the excessive complexity that increases process errors. Zero-shot CoT’s high variability (Q1 at 30% vs. median at 45%) exposes its fundamental limitation: the generic "step-by-step" instruction fails to adapt to domain-specific reasoning requirements. For instance, mathematical tasks require intermediate variable tracking and equation manipulation (e.g., solving for unknowns across multiple steps), while deductive reasoning benefits from direct premise-to-conclusion mapping without intermediate computations. This mismatch between CoT’s uniform step-wise approach and domain-specific cognitive requirements explains its unstable performance across reasoning types.

The systematic performance decline from SD1 to SD4 (from 62.26% to 47.26% in median accuracy) reveals a critical threshold effect in decomposition granularity. SD2-SD4’s simplified instructions lack essential structural elements—particularly modal operators and explicit task terminology—that prevent models from maintaining reasoning coherence during problem breakdown. This explains why SD2-SD4 fail: they trigger decomposition without providing the scaffolding necessary for maintaining logical consistency; instead, SD2-SD4 introduce extra information in the prompt that is meaningless for reasoning.

4.2 Token-Accuracy Paradox: More Computational Resources Don’t Guarantee Better Performance

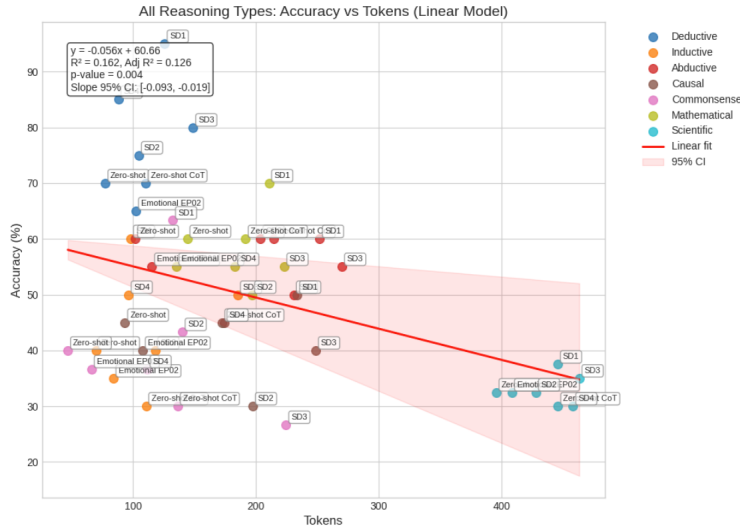


Figure 3: Overall reasoning accuracy vs. tokens.

Computational resource allocation paradoxically harms reasoning performance: higher token consumption correlates with lower accuracy ($R^2 = 0.162$, $p = 0.004$, slope = -0.056). This negative correlation directly contradicts the intuition that more detailed processing leads to better results—each additional token decreases accuracy by 0.056%, with statistical significance despite weak overall explanatory power (16% variance).

This paradox emerges from the mechanics of decomposition-induced error accumulation. Increased token usage typically results from excessive reasoning steps that introduce process mistakes (PM), as shown by the strong positive correlation between tokens and PM errors ($R^2 = 0.405$, $p < 0.001$). While longer outputs reduce wrong selection errors (WS) through better fact identification, they simultaneously create more opportunities for calculation mistakes, logical inconsistencies, and coherence failures during multi-step reasoning chains.

SD1 instances span a wide token range (100-350 tokens) while maintaining consistently low PM error rates (< 0.3), demonstrating unique resilience to the typical token-PM error correlation. In contrast, SD2-SD4 methods exhibit the expected positive correlation more dramatically, with PM errors escalating substantially at higher token counts. This differential response reveals that SD1’s

prompt design includes inherent safeguards that prevent decomposition-induced coherence failures. Mathematical reasoning demonstrates this clearly: as shown in Figure 4, SD1 maintains PM errors below 15% despite increased complexity, while SD2-SD4 show exponential PM growth (from 30% to 35%) due to insufficient structural scaffolding. This reveals why minimal prompting fails systematically: without explicit semantic constraints, models cannot distinguish between productive decomposition (breaking coherent logical units) and counter-productive fragmentation (arbitrary text splitting). The 30% performance gap between SD1 and SD4 in mathematical tasks directly quantifies the cost of removing structural guidance mechanisms.

4.3 WS-PM Error Trade-off Explains Decomposition Performance Variations

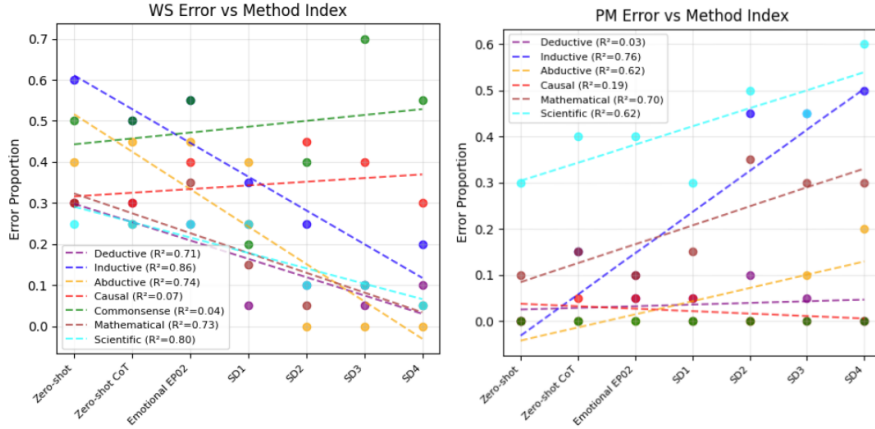


Figure 4: WS and PPM error rate vs. prompting methods.

As shown in Figure 4, SD variants systematically transform error patterns through a fundamental trade-off: decomposition reduces wrong selection (WS) errors but increases process mistakes (PM), with WS and PM errors constituting approximately 91% of all failures (WS: 61%, PM: 30%). Critically, SD variants differ in how well they manage this trade-off: SD1 maintains superior accuracy by reducing WS errors without significantly increasing PM errors, while SD2-SD4 show declining performance as PM errors overwhelm the WS improvements.

WS error reduction follows consistent patterns across most reasoning types as methods progress from zero-shot to SD variants. Deductive reasoning shows the strongest WS improvement ($R^2 = 0.71$, $p < 0.05$), declining from approximately 30% to 5%. Inductive, mathematical, and scientific reasoning achieve similar WS reductions ($R = 0.74 - 0.86$), while commonsense reasoning exhibits only WS errors throughout all methods, appearing solely in the WS analysis subplot of Figure 4. Notably, abductive reasoning presents an exception with increasing WS errors for SD1-SD4 variants, a phenomenon detailed in Section 4.4.

However, WS reduction comes at a cost: PM errors increase significantly with decomposition. Inductive reasoning presents the starkest trade-off ($R^2 = 0.76$), with PM errors escalating from zero values for baseline methods to 50% for SD4. Mathematical and scientific reasoning show dramatic increases ($R^2 = 0.70$ and 0.62 respectively), with PM errors rising from approximately 10% for zero-shot to 35% for SD2 in mathematical reasoning and from 30% to 60% for SD4 in scientific reasoning.

SD1 achieves optimal balance by reducing WS errors without significantly increasing other error types, particularly PM errors. In contrast, SD2-SD4 show a clear trade-off pattern: while successfully reducing WS errors, they introduce substantial PM errors, indicating that simplification breaks the delicate balance required for effective decomposition. This trade-off creates the inverse relationship between WS and PM errors observed across reasoning types, with different domains showing varying susceptibility to process errors.

As illustrated in Figure 5, WS errors demonstrate a statistically significant negative correlation with token consumption ($R^2 = 0.131$, $p = 0.011$), suggesting that longer responses tend to achieve better fact selection. Scientific reasoning tasks consistently cluster in the high-token, low-error region (400+ tokens, WS < 0.3), which reflects their inherent complexity that naturally requires more elaborate

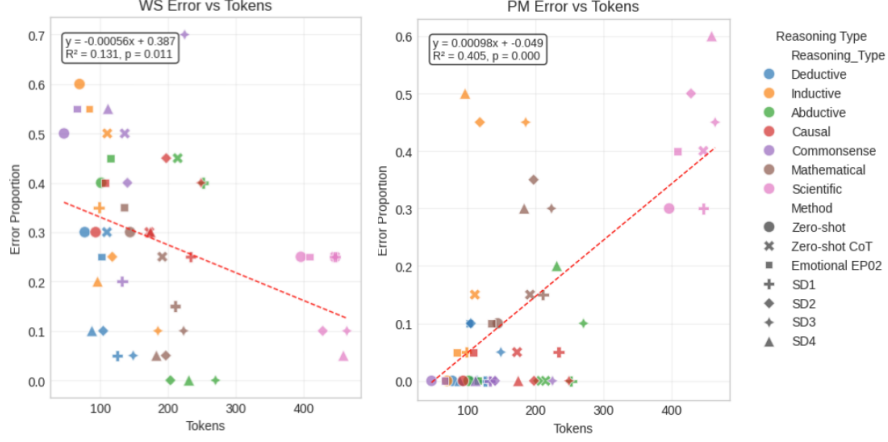


Figure 5: WS and PM error rate vs. tokens.

reasoning processes.

Conversely, PM errors exhibit a strong positive correlation with token consumption ($R^2 = 0.405$, $p < 0.001$, regression equation $y = 0.000998x + 0.049$), revealing that increased verbosity paradoxically in the response introduces more process mistakes. The stronger R^2 value (0.405 vs 0.131) indicates that token consumption better predicts PM errors than WS errors, demonstrating a stronger quantitative relationship between response length and process mistakes.

SD1 instances span a wide token range (100-350 tokens) while maintaining consistently low PM error rates (< 0.2), demonstrating unique resilience to the typical token-PM error correlation. In contrast, SD2-SD4 methods exhibit the expected positive correlation more dramatically, with PM errors escalating substantially at higher token counts. This differential response reveals that SD1’s structured approach maintains coherence across varying response lengths, enabling detailed analysis without the typical increase in process mistakes.

4.4 Abductive Reasoning Shows Critical Vulnerability to Decomposition Strategies

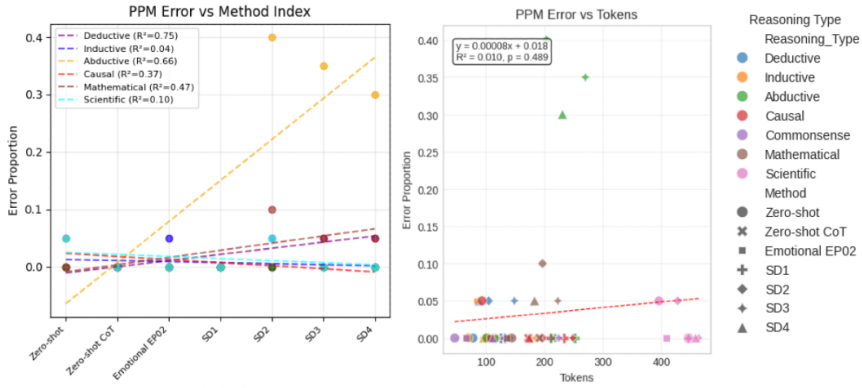


Figure 6: PPM error rate vs. prompting methods and tokens.

As demonstrated in Figure 6, Perspective Mistakes (PPM) exhibit extreme domain selectivity, with abductive reasoning showing catastrophic sensitivity to decomposition: PPM errors escalate from near-zero in zero-shot to 0.3-0.4 in SD2-SD4 ($R^2 = 0.66$). This dramatic vulnerability contrasts sharply with other reasoning types, where PPM errors remain negligible (< 0.1) regardless of decomposition strategy. The concentration of PPM errors in abductive reasoning reveals that inference-to-best-explanation tasks uniquely suffer when decomposition disrupts their inherent reasoning structure. PPM errors demonstrate no correlation with token consumption ($R^2 = 0.010$, $p = 0.489$), fundamentally differing from WS and PM errors’ token-dependent patterns. Most PPM instances cluster at near-zero error rates across all token ranges, with only abductive reasoning showing elevated values.

257 This independence demonstrates that PPM errors depend on decomposition structure rather than
 258 response verbosity, showing that the varying token counts produced by different SD methods have no
 259 impact on PPM rates.

260 Despite low absolute PPM rates in most domains, deductive ($R^2 = 0.75$) and mathematical reasoning
 261 ($R^2 = 0.47$) show strong method-dependent patterns while maintaining low error rates, indicat-
 262 ing sensitivity to decomposition approaches without severe consequences. Abductive reasoning
 263 presents a different profile: significant method dependence ($R^2 = 0.66$) combined with elevated
 264 error rates, making it both sensitive to method changes and prone to perspective mistakes. PPM
 265 error concentration follows an inverse relationship with task structure clarity. Deductive reasoning’s
 266 clear premise-conclusion structure naturally maps to sequential processing, enabling successful
 267 decomposition even with minimal guidance. Conversely, abductive reasoning requires simultaneous
 268 maintenance of competing hypotheses—a cognitive architecture that decomposition disrupts by
 269 forcing sequential evaluation of what should be parallel processes. The PPM difference between
 270 abductive and deductive reasoning under SD2-SD4 quantifies this structural incompatibility, demon-
 271 strating that effective decomposition must respect task-specific cognitive architectures rather than
 272 imposing uniform sequential processing.

273 The PPM analysis reveals fundamental limits of universal decomposition approaches. Abductive
 274 reasoning’s unique vulnerability, combined with its independence from token consumption patterns,
 275 suggests that one-size-fits-all prompting strategies will systematically fail for certain reasoning types.
 276 Future decomposition frameworks should incorporate domain-specific guards against PPM errors,
 277 particularly for inference tasks requiring simultaneous consideration of multiple explanatory hypothe-
 278 ses.

279 This selective vulnerability challenges assumptions about decomposition effectiveness across reason-
 280 ing domains. The concentration of PPM errors in abductive reasoning, despite its structural simplicity
 281 compared to mathematical or scientific tasks, suggests that task difficulty and decomposition com-
 282 patibility are orthogonal dimensions. This finding has profound implications for the development
 283 of general-purpose reasoning systems, indicating that effective prompting requires domain-aware
 284 architectural considerations beyond simple instruction complexity.

285 5 Limitations

286 While this investigation demonstrates SD1’s potential for enhancing autonomous problem-solving ca-
 287 pabilities, with notable improvements in deductive reasoning (95% accuracy on deductive reasoning)
 288 and up to 25% gains over baseline approaches, several important limitations warrant consideration.
 289 These constraints contextualize our findings and highlight opportunities for future research.

290 The error analysis reveals domain-specific limitations, particularly in PPM error patterns for abductive
 291 reasoning and the systematic trade-off between WS and PM errors. While SD1 maintains an optimal
 292 balance, the substantial performance decline in SD2-SD4 variants suggests that decomposition strate-
 293 gies are sensitive to structural prompt modifications. Moreover, the negative correlation between
 294 token consumption and overall accuracy ($R^2 = 0.162$, $p = 0.004$) indicates that computational
 295 efficiency does not uniformly translate to performance gains across reasoning types.

296 Our analysis reveals a fundamental tension where increased token consumption reduces WS errors
 297 but increases PM errors. This relationship suggests inherent limitations in current decomposition
 298 frameworks, where the very mechanisms designed to improve fact selection create conditions fa-
 299 voring process mistakes. The high variance in performance across reasoning types indicates that
 300 domain-specific optimizations may be necessary rather than universal decomposition strategies.

301 The performance ceiling observed in complex domains like scientific reasoning (37.5% maximum
 302 accuracy) suggests that SD strategies remain bounded by the underlying LLMs’ knowledge and
 303 architectural limitations. The inconsistent patterns in abductive reasoning across datasets (e.g., SD2’s
 304 100% accuracy on ART versus poor performance of 20% accuracy on α NLI) further highlight the
 305 limitations of current decomposition approaches for certain reasoning structures.

306 These limitations suggest several promising avenues: investigating domain-adaptive decomposition
 307 strategies that balance WS-PM error trade-offs, developing hybrid approaches for reasoning types
 308 sensitive to PPM errors, and exploring the integration of SD methods with other prompting techniques.
 309 Additionally, robustness testing under varied real-world conditions—including input noise, domain
 310 variations, and assumption violations—represents a critical area for future investigation.

6 Conclusion

The Self-Directed Decomposition (SD) methodology enhances LLMs’ autonomous reasoning capabilities by instructing models to decompose complex problems independently, without relying on human intervention or exemplar-based prompting. Unlike previous approaches that constrained models to predefined problem-solving trajectories, SD enables models to autonomously develop problem-solving strategies based on their understanding of tasks.

Experimental results across seven reasoning domains reveal that SD1 achieves superior performance by maintaining an optimal balance between error types, while subsequent variants (SD2-SD4) exhibit deteriorating performance despite reduced structural complexity. The comprehensive error analysis uncovers a fundamental trade-off in decomposition reasoning: while SD strategies systematically reduce wrong selection (WS) errors through improved fact selection, they simultaneously introduce process mistakes (PM) that increase with computational resource allocation. SD1’s exceptional performance stems from its resilience to this token-error paradox, maintaining consistent error control across varying token consumption levels.

The investigation further reveals domain-specific error patterns, particularly the concentration of perspective mistakes (PPM) in abductive reasoning, highlighting the limitations of universal decomposition prompting strategies. The negative correlation between overall token consumption and accuracy ($R^2 = 0.162, p = 0.004$) demonstrates that increased computational resources do not uniformly improve performance, challenging the intuitive assumption that more detailed processing leads to better results—a relationship our findings directly contradict.

These findings contribute to a nuanced understanding of decomposition-based prompting, revealing both its transformative potential and inherent limitations. The work demonstrates that effective prompt engineering requires careful calibration between methodological sophistication and error control, with different reasoning types demanding specialized optimization strategies. Future research should focus on developing adaptive frameworks that can dynamically adjust decomposition complexity based on domain characteristics, error risk profiles, and the fundamental token-error trade-offs identified in this investigation.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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